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$$f(x) = \ln(\sin(x) \cdot \sin(2x) \cdot \sin(3x))$$
$$\Rightarrow f'(x) = \frac{\cos(x) \cdot \sin(2x) \cdot \sin(3x) + \sin(x) \cdot 2 \cdot \cos(2x) \cdot \sin(3x) + \sin(x) + \cdot \sin(2x) \cdot 3 \cdot \cos(3x)}{\sin(x) \cdot \sin(2x) \cdot \sin(3x)}$$
$$= \frac{\cos(x)}{\sin(x)} + 2 \cdot \frac{\cos(2x)}{\sin(2x)} + 3 \cdot \frac{\cos(3x)}{\sin(3x)}$$
$$= \cot(x) + 2\cot(2x) + 3\cot(3x)$$

References